Bicycle Rental System – A Web-Based Bicycle Rental Platform

Synopsis

Introduction

Bicycle Rental System is proposed as a web-based application that enables users to find, reserve, rent, and return bicycles from registered rental points in a convenient and organised manner. The platform will display bicycle availability, rental details, booking status, and rental history in one place. It is designed to promote affordable transportation, support short-distance travel, and make bicycle rental operations more efficient for both customers and rental operators.

Problem Statement

- Customers often have difficulty finding nearby rental bicycles and knowing whether a bicycle is available.
- Manual booking registers and phone-based reservations can cause duplicate bookings and inaccurate records.
- Users may not receive clear information about rental rates, bicycle condition, pickup locations, or return deadlines.
- Rental operators lack a centralised system to manage bicycles, reservations, customers, and returned bicycles.
- There is limited feedback and maintenance tracking, making it difficult to maintain service quality and bicycle safety.

Requirements

Software Requirements

1. Frontend: React.js (JavaScript)
2. Backend: Python (FastAPI)
3. Database: MongoDB
4. Development Tools: VS Code, Git, Postman
5. Deployment: Vercel (frontend), Render (backend), MongoDB Atlas (database)

Scope of Project

The initial scope of the Bicycle Rental System is limited to a web-based application that supports customer registration, bicycle browsing, availability checking, online reservation requests, rental-status tracking, and bicycle inventory management. Rental operators or administrators will be able to add bicycles, update availability, confirm bookings, record returns, and mark bicycles for maintenance. During the initial phase, payments will be handled at the rental counter or through an external method, and the system will not include live GPS tracking, smart-lock integration, or a dedicated mobile application. These features may be considered in later phases based on user feedback and operational requirements.

Features

- Searchable bicycle catalogue showing bicycle type, rental rate, condition, and pickup location.
- Availability calendar that helps users select a suitable date and rental duration.
- Reservation system allowing customers to request, confirm, cancel, and track bicycle bookings.
- Customer profile with contact details, active rentals, previous bookings, and rental history.
- Operator dashboard for adding bicycles, updating status, managing reservations, and recording returns.
- Rental status updates such as available, reserved, rented, returned, and under maintenance.
- Feedback and rating system to help improve bicycle quality and customer service.
- Notification and reporting options for booking updates, overdue returns, damaged bicycles, and service issues.

Modules

- **User Management Module:** Handles registration, login, profile management, and role-based access for customers and operators.
- **Bicycle Inventory Module:** Stores bicycle details, categories, rental rates, locations, availability, and maintenance status.
- **Search and Availability Module:** Allows users to search bicycles by location, type, price, and date, and view current availability.
- **Booking and Rental Module:** Manages reservation requests, booking confirmation, pickup details, return deadlines, and rental status.
- **Maintenance Module:** Records inspections, repairs, reported damage, and bicycle maintenance history.
- **Feedback and Reporting Module:** Allows ratings, reviews, issue reporting, and administrative review of suspicious activity.

Type of Users

- **Primary Users:** Students, commuters, tourists, and local residents who need bicycles for short-distance travel.
- **Secondary Users:** Bicycle rental operators or staff members who manage bicycles, reservations, pickups, and returns.
- **Administrator:** Maintains the platform, manages users and rental locations, monitors reports, and oversees system data.
- **Future Users:** Delivery partners, campus transport departments, and bicycle-sharing organisations that may be integrated in later phases.

Future Scope of Improvement

- Integration of an online payment gateway for secure rental payments, deposits, and refunds.
- Live GPS tracking, route support, and map-based discovery of bicycles and rental stations.
- Dedicated mobile applications for Android and iOS with push notifications.
- Smart locks, QR-code unlocking, and IoT sensors for automated bicycle access and monitoring.
- Dynamic pricing, subscription plans, loyalty rewards, and membership packages.
- Predictive maintenance, usage analytics, and demand-based bicycle placement.

Conclusion

The Bicycle Rental System addresses the challenges of finding and managing rental bicycles by bringing customers, bicycles, and rental operators onto one organised platform. It will make bicycle availability and booking information easier to access while reducing manual record-keeping and booking errors. By introducing the core features first and expanding the system according to actual usage, the project aims to provide a practical, scalable, and user-friendly solution that encourages affordable and eco-friendly transportation.

Declaration

I hereby declare that the synopsis titled “Bicycle Rental System – A Web-Based Bicycle Rental Platform” is an original concept developed by me, based on identified transportation needs and practical feasibility. The information presented in this document is true and accurate to the best of my knowledge.

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